Calculus II

Tutorial Problems #1 17 – 21 Feb 2025

◇ **Topics:**

- Functions of several variables
- Level curve, level surface
- Limit

In Problems 1–3, find and sketch the domain for each function.

Problem 1. $f(x, y) = \sqrt{y - x - 2}$.

Problem 2. $f(x, y) = \dfrac{(x-1)(y+2)}{(y-x)(y-x^3)}$.

Problem 3. $f(x, y) = \sqrt{(x^2-4)(y^2-9)}$.

In Problems 4–5, display the values of the functions in two ways:

(a) by sketching the surface $z = f(x, y)$

(b) by drawing an assortment of level curves in the function's domain. Label each level curve with its function value.

Problem 4. $f(x, y) = 1 - |y|$.

Problem 5. $f(x, y) = \sqrt{x^2 + y^2 + 4}$.

In Problems 6–7, sketch a typical level surface for the function.

Problem 6. $f(x, y, z) = x^2 + y^2 + z^2$.

Problem 7. $f(x, y, z) = z - x^2 - y^2$.

Problem 8. Find the limits of a function of two variables

$$\lim_{(x,y)\to(0,\pi/4)} \sec x \tan y.$$

In Problems 9–10, find the limit by rewriting the fractions first.

Problem 9.

$$\lim_{(x,y)\to(0,0)} \frac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}.$$

Problem 10.

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2}.$$

Problem 11. Find the limits of a function of three variables

$$\lim_{P\to(\pi,\pi,0)} (\sin^2 x + \cos^2 y + \sec^2 z).$$

END

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Answers to selected problems

1. $y \geq x+2$
2. $y \neq x$ and $y \neq x^3$
3. Solution of $(x^2-4)(y^2-9) \geq 0$.
8. 1
9. 1
10. 2
11. 2